

Rational Expressions & Equations — Solutions

Complete Step-by-Step Solutions with Trap Alerts & Key Insights

Question 1 [Easy]

Correct Answer: A) $x - 3$

Step-by-Step Solution:

- $x^2 - 9 = (x - 3)(x + 3)$.
- $(x - 3)(x + 3) / (x + 3) = x - 3$ (for $x \neq -3$).

Why each wrong option fails:

- B) $x + 3$ — Cancelled the wrong factor.
- C) $x^2 - 3$ — Subtracted instead of factoring.
- D) $(x-3)(x+3)$ — This is the numerator before cancelling.

■ ■ **TRAP ALERT:** Students must recognize $x^2 - 9$ as a difference of squares.

■ **KEY INSIGHT:** $a^2 - b^2 = (a-b)(a+b)$. Cancel the common factor.

■ **FAST METHOD:** Factor, cancel. Two steps.

■ 🏠 **VERDICT:** Hand-solve — factoring is faster than any tool.

Question 2 [Easy]

Correct Answer: A) $3x^2$

Step-by-Step Solution:

- $6x^3 / 2x = (6/2)(x^3/x) = 3x^2$.

Why each wrong option fails:

- B) $4x^2$ — Subtracted coefficients (6-2) instead of dividing.
- C) $3x^3$ — Divided coefficients but forgot to reduce the exponent.
- D) $6x^2$ — Only reduced the exponent, not the coefficient.

■ ■ **TRAP ALERT:** When dividing monomials: divide coefficients AND subtract exponents.

■ **KEY INSIGHT:** $x^3/x = x^{3-1} = x^2$.

■ **FAST METHOD:** $6/2 = 3$, $x^3/x = x^2 \rightarrow 3x^2$.

■ 🏠 **VERDICT:** Hand-solve — basic exponent rules.

Question 3 [Easy]

Correct Answer: B) 2

Step-by-Step Solution:

- A fraction is undefined when the denominator = 0.
- $x - 2 = 0 \rightarrow x = 2$.

Why each wrong option fails:

- A) -5 — This makes the numerator 0 (the expression equals 0, not undefined).
- C) -2 — Sign error on the denominator.
- D) 5 — Confused numerator and denominator.

■ TRAP ALERT: Setting the NUMERATOR = 0 instead of the denominator is a classic error.

■ KEY INSIGHT: Undefined = denominator is 0. Zero = numerator is 0 (but denominator isn't).

■ FAST METHOD: Denominator = 0 $\rightarrow x - 2 = 0 \rightarrow x = 2$.

■ VERDICT: Hand-solve — conceptual.

Question 4 [Easy]

Correct Answer: A) $5/(x + 1)$

Step-by-Step Solution:

- Same denominator: $3/(x+1) + 2/(x+1) = (3+2)/(x+1) = 5/(x+1)$.

Why each wrong option fails:

- B) $5/(2x+2)$ — Added the denominators.
- C) $6/(x+1)^2$ — Multiplied numerators and denominators.
- D) $5/(x+1)^2$ — Squared the denominator.

■ TRAP ALERT: When denominators are the same, keep the denominator and add numerators only.

■ KEY INSIGHT: Same denominator $\rightarrow$ add numerators only. Never add denominators.

■ FAST METHOD: Same denom $\rightarrow (3+2)/(x+1) = 5/(x+1)$.

■ VERDICT: Hand-solve — basic fraction rule.

Question 5 [Medium]

Correct Answer: A) $(x + 2)/(x + 6)$

Step-by-Step Solution:

- Numerator: $x^2 - 4x - 12 = (x - 6)(x + 2)$.
- Denominator: $x^2 - 36 = (x - 6)(x + 6)$.
- Cancel $(x - 6)$: $(x + 2)/(x + 6)$.

Why each wrong option fails:

- B) $(x-2)/(x-6)$ — Wrong factoring of numerator.
- C) $(x+2)/(x-6)$ — Cancelled the wrong factor.
- D) $(x-6)/(x+2)$ — Inverted the fraction.

■ ■ **TRAP ALERT:** Finding two numbers that multiply to -12 and add to -4: that's -6 and +2.

■ **KEY INSIGHT:** Factor both, cancel common factors. Always check by multiplying back.

■ **FAST METHOD:** Factor → cancel. Recognize difference of squares in the denominator.

■ 🏠 **VERDICT:** Hand-solve — systematic factoring.

Question 6 [Easy]

Correct Answer: 3.5

Step-by-Step Solution:

- $5/(x - 3) = 10$
- $5 = 10(x - 3)$
- $5 = 10x - 30$
- $35 = 10x$
- $x = 3.5$ (or $7/2$)

■ ■ **TRAP ALERT:** Forgetting to distribute the 10 to both terms: $10(x)$ AND $10(-3)$.

■ **KEY INSIGHT:** Cross-multiply, then solve the linear equation.

■ **FAST METHOD:** $5/10 = x - 3 \rightarrow 1/2 = x - 3 \rightarrow x = 7/2$.

■ 🏠 **VERDICT:** Hand-solve — one-step proportion.

Question 7 [Medium]

Correct Answer: A) $2/(x + 2)$

Step-by-Step Solution:

- Numerator: $2x + 6 = 2(x + 3)$.
- Denominator: $x^2 + 5x + 6 = (x + 2)(x + 3)$.
- Cancel $(x + 3)$: $2/(x + 2)$.

Why each wrong option fails:

- B) $2/(x + 3)$ — Cancelled the wrong factor.
- C) $(2x+6)/(x+2)$ — Only factored the denominator.
- D) $2x/(x+2)$ — Cancelled only part of the numerator.

■ TRAP ALERT: Factor the numerator by pulling out the GCF (2), then cancel.

■ KEY INSIGHT: Always factor the GCF from the numerator before looking for common factors.

■ FAST METHOD: $2(x+3)/[(x+2)(x+3)] = 2/(x+2)$.

■ VERDICT: Hand-solve — GCF + factor + cancel.

Question 8 [Medium]

Correct Answer: A) 1

Step-by-Step Solution:

- $4/x + 3/x = 7/x = 7$.
- $7/x = 7 \rightarrow x = 1$.

Why each wrong option fails:

- B) 7 — Set $x = 7$ by confusing numerator and denominator.
- C) $7/4$ — Only used the first fraction.
- D) $1/7$ — Inverted the correct answer.

■ TRAP ALERT: Combine like fractions first: $4/x + 3/x = 7/x$, THEN solve.

■ KEY INSIGHT: Same denominator $\rightarrow$ add numerators first.

■ FAST METHOD: $7/x = 7 \rightarrow x = 1$.

■ VERDICT: Hand-solve — combine then solve.

Question 9 [Medium]

Correct Answer: A) $(x + 5)/(x - 2)$

Step-by-Step Solution:

- Numerator: $x^2 + 7x + 10 = (x + 5)(x + 2)$.
- Denominator: $x^2 - 4 = (x - 2)(x + 2)$.
- Cancel $(x + 2)$: $(x + 5)/(x - 2)$.

Why each wrong option fails:

- B) $(x+5)/(x+2)$ — Cancelled $(x-2)$ instead of $(x+2)$.
- C) $(x+2)/(x-2)$ — Cancelled $(x+5)$ which isn't common.
- D) $(x-5)/(x-2)$ — Wrong sign in factoring.

■ ■ **TRAP ALERT:** The common factor is $(x + 2)$, not $(x - 2)$.

■ **KEY INSIGHT:** Always factor completely before cancelling.

■ **FAST METHOD:** Factor both $\rightarrow$ cancel $(x+2) \rightarrow (x+5)/(x-2)$.

■ 🏠 **VERDICT:** Hand-solve — standard factoring.

Question 10 [Medium]

Correct Answer: B) 7

Step-by-Step Solution:

- $3/(x + 2) = 1/(x - 4)$
- Cross-multiply: $3(x - 4) = 1(x + 2)$
- $3x - 12 = x + 2$
- $2x = 14$
- $x = 7$
- Check: $3/9 = 1/3$ and $1/3 = 1/3$ ✓

Why each wrong option fails:

- A) -7 — Sign error when solving.
- C) 5 — Arithmetic error.
- D) -5 — Wrong setup.

■ ■ **TRAP ALERT:** Always check the solution doesn't make a denominator zero. $x = 7$ doesn't, so it's valid.

■ **KEY INSIGHT:** Cross-multiply rational equations, then solve the linear equation. Always check for extraneous solutions.

■ **FAST METHOD:** Cross-multiply $\rightarrow$ linear equation $\rightarrow$ solve.

■ 🏠 **VERDICT:** Hand-solve — cross-multiplication is fastest.

Question 11 [Medium]

Correct Answer: A) $x + 1$

Step-by-Step Solution:

- $(x^2 - 1)/(x + 2) \cdot (x + 2)/(x - 1)$
- $= (x^2 - 1)(x + 2) / [(x + 2)(x - 1)]$
- Cancel $(x + 2)$: $(x^2 - 1)/(x - 1)$
- Factor: $(x-1)(x+1)/(x-1)$
- Cancel $(x - 1)$: $x + 1$

Why each wrong option fails:

- B) $x - 1$ — Cancelled $(x+1)$ instead of $(x-1)$.
- C) $x^2 - 1$ — Forgot to cancel $(x - 1)$.
- D) $(x+1)(x+2)$ — Only cancelled one factor.

■ ■ TRAP ALERT: Must factor $x^2 - 1 = (x-1)(x+1)$ to cancel with $(x-1)$.

■ KEY INSIGHT: When multiplying fractions: multiply across, then cancel common factors.

■ FAST METHOD: Cancel $(x+2)$, factor x^2-1 , cancel $(x-1) \rightarrow x+1$.

■ 🏠 VERDICT: Hand-solve — cancel before multiplying.

Question 12 [Hard]

Correct Answer: 11/5

Step-by-Step Solution:

- LCD = $(x-1)(x+1) = x^2 - 1$.
- Multiply each term by LCD:
- $2(x+1) + 3(x-1) = 10$
- $2x + 2 + 3x - 3 = 10$
- $5x - 1 = 10$
- $5x = 11$
- $x = 11/5$
- Check: $x = 11/5 \neq \pm 1$, so it's valid.

■ ■ TRAP ALERT: $x = 1$ and $x = -1$ make the denominators zero, so they must be excluded.

■ KEY INSIGHT: Multiply every term by the LCD, then solve the resulting linear equation.

■ FAST METHOD: LCD = $(x-1)(x+1)$. Multiply through, solve, check for extraneous solutions.

■ 🏠 VERDICT: Hand-solve — LCD method is standard.

Question 13 [Hard]

Correct Answer: A) $(b - a)/(b + a)$

Step-by-Step Solution:

- Numerator: $1/a - 1/b = (b - a)/(ab)$.
- Denominator: $1/a + 1/b = (b + a)/(ab)$.
- Dividing: $[(b-a)/(ab)] / [(b+a)/(ab)] = (b-a)/(b+a)$.
- The ab cancels.

Why each wrong option fails:

- B) $(a-b)/(a+b)$ — Reversed the subtraction order.
- C) $ab/(a+b)$ — Incorrectly simplified.
- D) $(a+b)/ab$ — Inverted the result.

■ ■ **TRAP ALERT:** The key step is multiplying by the reciprocal when dividing complex fractions.

■ **KEY INSIGHT:** Complex fractions: find common denominator for top and bottom, then divide.

■ **FAST METHOD:** The ab denominators cancel when you divide. Just focus on numerators.

■ 🏠 **VERDICT:** Hand-solve — complex fraction simplification.

Question 14 [Hard]

Correct Answer: 7

Step-by-Step Solution:

- Numerator: $3x^2 - 12 = 3(x^2 - 4) = 3(x-2)(x+2)$.
- Denominator: $x^2 - 4x + 4 = (x - 2)^2$.
- Simplify: $3(x-2)(x+2)/(x-2)^2 = 3(x+2)/(x-2)$ (for $x \neq 2$).
- $f(5) = 3(5+2)/(5-2) = 3(7)/3 = 7$.

■ ■ **TRAP ALERT:** Must factor and simplify BEFORE substituting $x = 5$.

■ **KEY INSIGHT:** Always simplify the rational expression before evaluating.

■ **FAST METHOD:** Factor $\rightarrow$ cancel one $(x-2) \rightarrow 3(x+2)/(x-2) \rightarrow f(5) = 21/3 = 7$.

■ 🏠 **VERDICT:** Hand-solve — factor, cancel, evaluate.

Question 15 [Hard]

Correct Answer: A) 0

Step-by-Step Solution:

- Multiply both sides by $(x - 3)$:
- $x - 2(x - 3) = 3$
- $x - 2x + 6 = 3$
- $-x + 6 = 3$
- $x = 3$
- But $x = 3$ makes the denominator 0! So $x = 3$ is extraneous.
- There are NO valid solutions.

Why each wrong option fails:

- B) 1 — Would mean $x = 3$ is valid, but it makes denominators zero.
- C) 2 — The equation doesn't produce two solutions.
- D) Infinitely many — The equation is not an identity.

■ ■ TRAP ALERT: Solving algebraically gives $x = 3$, but this is **EXTRANEIOUS** (makes denominator = 0).

■ KEY INSIGHT: Always check rational equation solutions in the original denominators!

■ FAST METHOD: Solve → get $x = 3$ → check denominator → undefined → 0 solutions.

■ 🏠 VERDICT: Hand-solve — the key insight is checking for extraneous solutions.

Question 16 [Hard]

Correct Answer: A) $(2x + 5)/(x + 3)$

Step-by-Step Solution:

- Numerator: $4x^2 - 25 = (2x - 5)(2x + 5)$.
- Denominator: $2x^2 + x - 15$. Find factors: need $ac = -30$, $b = 1$.
- Factors: 6 and -5. Split: $2x^2 + 6x - 5x - 15 = 2x(x+3) - 5(x+3) = (2x-5)(x+3)$.
- Cancel $(2x - 5)$: $(2x + 5)/(x + 3)$.

Why each wrong option fails:

- B) $(2x-5)/(x-3)$ — Wrong signs in factoring.
- C) $(2x-5)/(x+3)$ — Cancelled $(2x+5)$ instead of $(2x-5)$.
- D) $(2x+5)/(x-3)$ — Wrong sign in denominator factoring.

■ ■ TRAP ALERT: Factoring the denominator (trinomial with leading coefficient $\neq 1$) requires the AC method.

■ KEY INSIGHT: Difference of squares (numerator) + AC method (denominator) → cancel common factor.

■ FAST METHOD: Factor both → cancel $(2x-5)$ → $(2x+5)/(x+3)$.

■ 🏠 VERDICT: Hand-solve — factoring practice.

Question 17 [Hard]

Correct Answer: B) $x^2 - 2x + 4$

Step-by-Step Solution:

- $x^3 + 8 = x^3 + 2^3 = (x + 2)(x^2 - 2x + 4)$.
- Divide by $(x + 2)$: $x^2 - 2x + 4$.

Why each wrong option fails:

- A) $x^2 + 4$ — Missing the $-2x$ middle term.
- C) $x^2 + 2x + 4$ — Wrong sign on the middle term.
- D) $x^2 - 4$ — Confused with difference of squares.

■ TRAP ALERT: Sum of cubes: $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$. The middle term is **NEGATIVE**.

■ KEY INSIGHT: Memorize: sum of cubes $\rightarrow (a+b)(a^2-ab+b^2)$. Difference of cubes $\rightarrow (a-b)(a^2+ab+b^2)$.

■ FAST METHOD: Recognize sum of cubes pattern instantly. $a=x, b=2$.

■ VERDICT: Hand-solve — pattern recognition (sum of cubes).

Question 18 [Hard]

Correct Answer: B) 2

Step-by-Step Solution:

- LCD = $(a - 3)(a + 3) = a^2 - 9$.
- Rewrite both left-side fractions with the LCD:
- $3(a + 3)/[(a - 3)(a + 3)] + B(a - 3)/[(a - 3)(a + 3)] = (5a + 3)/[(a - 3)(a + 3)]$.
- Since denominators are equal, set numerators equal:
- $3(a + 3) + B(a - 3) = 5a + 3$.
- Expand: $3a + 9 + Ba - 3B = 5a + 3$.
- Group by terms: $(3 + B)a + (9 - 3B) = 5a + 3$.
- Match the coefficient of a : $3 + B = 5$, so $B = 2$.
- Verify the constant term: $9 - 3(2) = 3$. This matches.

Why each wrong option fails:

- A) 1 — Incorrect coefficient matching.
- C) $a + 1$ — B must be a constant.
- D) $a + 3$ — B must be a constant.

■ TRAP ALERT: Match coefficients of both the a -term and the constant term to verify.

■ KEY INSIGHT: Partial fractions: equate numerators after getting the common denominator.

■ FAST METHOD: Coefficient of a gives B directly. Constant term verifies.

■ VERDICT: Hand-solve — partial fraction decomposition.

Question 19 [Hard]

Correct Answer: 4

Step-by-Step Solution:

- LCD = $(x + 2)(x - 2) = x^2 - 4$.
- Multiply every term by the LCD: $(x - 2) + (x + 2) = k$.
- Simplify: $2x = k$, so $x = k/2$.
- Excluded values: $x = 2$ and $x = -2$ make denominators zero.
- For NO solution, $x = k/2$ must be excluded.
- If $k = 4$: $x = 2$, which is excluded. No valid solution exists.
- Therefore $k = 4$.

■ TRAP ALERT: The equation simplifies to $2x = k$. No solution occurs when $x = k/2$ is an excluded value ($x = 2$ or $x = -2$).

■ KEY INSIGHT: No solution in rational equations often means the algebraic solution is extraneous.

■ FAST METHOD: Simplify $\rightarrow 2x = k \rightarrow x = k/2$. Excluded: $x = \pm 2$. So $k = 4$ or $k = -4$.

■ VERDICT: Hand-solve — simplify first, then identify extraneous conditions.

Question 20 [Medium]

Correct Answer: B) $5p/48$

Step-by-Step Solution:

- First half: $p/2$ problems at 8 problems/hour. Time = $(p/2)/8 = p/16$ hours.
- Second half: $p/2$ problems at 12 problems/hour. Time = $(p/2)/12 = p/24$ hours.
- Total time = $p/16 + p/24$.
- LCD = 48: $3p/48 + 2p/48 = 5p/48$.

Why each wrong option fails:

- A) $p/20$ — Averaged the rates (wrong approach).
- C) $p/4$ — Used only one rate.
- D) $20/p$ — Inverted the expression.

■ TRAP ALERT: You CANNOT average rates. Must find time for each part separately, then add.

■ KEY INSIGHT: Time = Work / Rate. Find time for each part, then add.

■ FAST METHOD: $p/16 + p/24 = p(3+2)/48 = 5p/48$.

■ VERDICT: Hand-solve — rate/time/work formula.

■ TOPIC SUMMARY

1. **UNDEFINED VALUES:** Set denominator = 0. These values are excluded from the domain.
2. **SIMPLIFYING:** Factor numerator and denominator, then cancel common factors.
3. **ADDING FRACTIONS:** Same denominator → add numerators. Different → find LCD first.
4. **MULTIPLYING FRACTIONS:** Multiply across. Cancel common factors before multiplying.
5. **DIVIDING FRACTIONS:** Multiply by the reciprocal of the divisor.
6. **SOLVING RATIONAL EQUATIONS:** Multiply by LCD → solve → CHECK for extraneous solutions!
7. **EXTRANEIOUS SOLUTIONS:** Solutions that make a denominator = 0 are invalid. Always check!
8. **DIFFERENCE OF SQUARES:** $a^2 - b^2 = (a-b)(a+b)$. Very common in rational expressions.
9. **SUM/DIFFERENCE OF CUBES:** $a^3 + b^3 = (a+b)(a^2-ab+b^2)$. Watch the middle term sign.
10. **DESMOS vs HAND:** Hand-solve for factoring and simplification. Desmos for graphing to find intersections and verify solution count.